

The Internal Structure of the Eye

The human eye is a roughly spherical organ about 24 millimetres across. It works like a camera: a clear front window and an adjustable lens focus light onto a light-sensitive sheet at the back, which converts the image into electrical signals for the brain.

The wall of the eyeball is built from three concentric layers, called tunics. The outer fibrous tunic (sclera and cornea) gives the eye its shape and protects it. The middle vascular tunic, or uvea (choroid, ciliary body and iris), carries blood vessels and controls light and focus. The inner neural tunic is the retina, where vision begins.

Inside these layers the eye is divided by the lens into two main segments. The anterior segment in front of the lens is filled with a watery fluid called aqueous humor. The larger posterior segment behind the lens is filled with a clear gel called the vitreous body.

Try it: select the paragraph about the three tunics and tap Summarize.

The outer layer: sclera and cornea

The sclera is the tough, white outer coat that covers about five sixths of the eyeball. It is made of densely packed collagen fibres arranged irregularly, which makes it strong but opaque. The six extraocular muscles that move the eye attach to the sclera, and at the back it is pierced by the optic nerve.

At the front, the sclera continues into the cornea, the transparent dome that covers the iris and pupil. The cornea has no blood vessels; it takes oxygen from the air and nutrients from tears and aqueous humor. Because of its curved shape it provides about two thirds of the eye's total focusing power.

The cornea has five classic layers: the epithelium, Bowman's layer, the stroma, Descemet's membrane and the endothelium. The stroma makes up about 90 percent of its thickness. The endothelium pumps water out of the stroma to keep it clear, and in humans these cells do not regenerate.

The ring where cornea meets sclera is the limbus. It holds the stem cells that renew the corneal surface and lies close to the drainage channels for aqueous humor.

The middle layer: the uvea

The choroid is a thin, dark, highly vascular layer between the sclera and the retina. Its blood vessels nourish the outer part of the retina, including the photoreceptors, and its pigment absorbs stray light so that it does not scatter inside the eye.

Toward the front, the choroid thickens into the ciliary body. It contains the ciliary muscle, which is connected to the lens by fine suspensory fibres called zonules. Folds on its inner surface, the ciliary processes, secrete aqueous humor.

The iris is the coloured ring at the front of the uvea, with the pupil at its centre. Two smooth muscles control pupil size. The circular sphincter pupillae, supplied by parasympathetic nerves, constricts the pupil in bright light. The radial dilator pupillae, supplied by sympathetic nerves, widens it in dim light. Eye colour depends on the amount of melanin in the iris.

Chambers, fluids and the lens

The space in front of the lens is split by the iris into an anterior chamber (between cornea and iris) and a posterior chamber (between iris and lens). Aqueous humor is made by the ciliary processes in the posterior chamber, flows through the pupil into the anterior chamber, and drains through the trabecular meshwork into the canal of Schlemm.

The balance between production and drainage sets the intraocular pressure, normally about 10 to 21 mmHg. If drainage is impaired the pressure can rise and damage the optic nerve, a condition known as glaucoma.

The lens is a transparent, biconvex structure with no blood vessels or nerves, made of tightly packed cells filled with proteins called crystallins. To focus on a near object, the ciliary muscle contracts, the zonules slacken, and the elastic lens becomes rounder. This is called accommodation. With age the lens stiffens (presbyopia) and can become cloudy (cataract).

Behind the lens, the vitreous body fills about 80 percent of the eye's volume. It is roughly 99 percent water held in a mesh of collagen and hyaluronic acid, and it helps keep the retina pressed against the choroid.

The retina: where vision begins

The retina lines the back of the eye and has about ten microscopic layers. Its outermost layer, the retinal pigment epithelium, supports the photoreceptors and recycles their visual pigments. Oddly, the photoreceptors face away from the incoming light, which must pass through the other retinal layers first.

There are two kinds of photoreceptor. About 120 million rods are very sensitive to light and handle night and peripheral vision, but do not detect colour. About 6 million cones work in bright light and provide sharp, colour vision through three types tuned to short, medium and long wavelengths.

Signals pass from photoreceptors to bipolar cells and then to ganglion cells, while horizontal and amacrine cells connect neighbouring pathways sideways. The axons of roughly 1.2 million ganglion cells gather at the optic disc and leave the eye as the optic nerve.

The macula is a small yellowish area near the centre of the retina. At its centre, the fovea contains densely packed cones and gives the sharpest vision. The optic disc has no photoreceptors at all, which creates a blind spot in each eye that the brain fills in.

From eye to brain, and quick review

Each optic nerve runs back to the optic chiasm, where fibres from the nasal half of each retina cross to the other side. The optic tracts carry the signals to the lateral geniculate nucleus of the thalamus, and the optic radiations relay them to the primary visual cortex in the occipital lobe.

Quick review. Sclera: white protective coat. Cornea: clear window, most of the focusing power. Choroid: blood supply to the outer retina. Ciliary body: makes aqueous humor and changes lens shape. Iris: controls pupil size. Lens: fine focusing. Vitreous: gel that holds the eye's shape. Retina: converts light into nerve signals. Optic nerve: carries them to the brain.

Try it: select the quick review above and tap Flashcards.

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